Exam

Name___________________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Digestion refers to the 1) ________
A) input of food into the digestive tract.
B) release of substances into the lumen of the gut.
C) breakdown of food into particles small enough to cross epithelial cells.
D) absorption of nutrients in the gut.
E) progressive dehydration of indigestible residue.

2) Secretion refers to the 2) ________
A) progressive dehydration of indigestible residue.
B) breakdown of food into particles small enough to cross epithelial cells.
C) release of substances into the lumen of the gut.
D) absorption of nutrients in the gut.
E) input of food into the digestive tract.

3) Which are most closely regulated by the body? 3) ________
A) motility, secretion, and digestion
B) digestion and motility
C) secretion and digestion
D) absorption and motility
E) motility and secretion

4) Which is an accessory organ of digestion? 4) ________
A) esophagus B) colon C) pancreas D) stomach E) spleen

5) The ________ is a significant site of absorption of water and electrolytes, but NOT of nutrients. 5) ________
A) mouth
B) small intestine
C) stomach
D) large intestine
E) None of the answers are correct.

6) Which is the sequence of layers from the lumen to the outer wall of the digestive tract? 6) ________
A) serosa, submucosa, mucosa, muscularis externa
B) submucosa, mucosa, serosa, muscularis externa
C) mucosa, submucosa, muscularis externa, serosa
D) submucosa, muscularis externa, serosa, mucosa
E) mucosa, submucosa, serosa, muscularis externa

7) The lamina propria and mucous epithelium are parts of the 7) ________
A) serosa.
B) submucosa.
C) mucosa.
D) muscularis mucosa.
E) adventitia.

8) The layer of connective tissue between the digestive epithelium and a layer of smooth muscle is the 8) ________
A) submucosa.
B) lamina propria.
C) myenteric plexus.
D) muscularis mucosae.
E) submucosal plexus.

9) Contraction of the ________ alters the surface area by moving villi. 9) ________
A) submucosal plexus
B) adventitia
C) submucosa
D) muscularis mucosae
E) mucosa

10) Rugae, plicae, and villi are all structures that 10) _______
A) increase surface area.
B) secrete hormones.
C) provide mechanical digestion.
D) secrete enzymes.
E) provide immunity.

11) The motor activity of the muscularis externa is controlled by the 11) _______
A) motilin.
B) extrinsic neurons.
C) submucosal plexus.
D) migrating motor complex.
E) myenteric plexus.

12) Intestinal crypts 12) _______
A) only produce new cells for the mucosa of the small intestine.
B) only increase the surface area of the mucosa of the small intestine.
C) only function in the absorption of nutrients.
D) increase the surface area of the mucosa of the small intestine and produce new cells for the mucosa of the small intestine.
E) only carry products of digestion that will not pass through the walls of blood capillaries.

13) Chief cells secrete 13) _______
A) mucus.
B) pepsinogen.
C) gastrin.
D) intrinsic factor.
E) hydrochloric acid.

14) G cells of the stomach secrete 14) _______
A) pepsin.
B) enterokinase.
C) secretin.
D) cholecystokinin.
E) gastrin.

15) In the intestine, the epithelial cells have 15) ________
A) electrical gap junctions. B) tight junctions. C) leaky junctions.

16) The subepithelial connective tissue of the GI tract, immediately beneath the epithelium, is the 16) ________
A) muscularis mucosae.
B) submucosa.
C) submucosal plexus.
D) serosa.
E) lamina propria.

17) Between the layer of circular and longitudinal muscle in the muscularis externa is the 17) ________
A) muscularis mucosa.
B) submucosa.
C) submucosal plexus.
D) mucosa.
E) myenteric plexus.

18) The myenteric plexus is 18) ________
A) primarily composed of connective tissue.
B) a network of nerves.
C) a layer of longitudinal smooth muscle.
D) a layer of circular smooth muscle.
E) the mucus secreting layer of the digestive tract.

19) Peyer's patches are found in the 19) ________
A) stomach.
B) pancreas.
C) esophagus.
D) intestine.
E) colon.

20) Features of the submucosa include 20) ________
A) blood vessels, lymph vessels, and a major nerve network.
B) a major nerve network.
C) blood and lymph vessels.
D) Peyer's patches.
E) blood vessels, lymph vessels, a major nerve network, and Peyer's patches.

21) The ________ are sheets of peritoneal membrane that hold some of the intestines in their place. 21) ________
A) adventitia
B) serosa
C) lamina propria
D) fibrosa
E) mesenteries

22) Slow waves are 22) ________
A) segmental contractions.
B) reflexes that originate and are integrated in the enteric nervous system.
C) cycles of smooth muscle contraction and relaxation.
D) peristaltic contractions.
E) cycles of depolarization and repolarization.

23) ________ are pacemakers for slow wave activity. 23) _______
A) Interstitial cells of Cajal
B) G cells
C) Intrinsic neuron cells
D) Extrinsic neuron cells
E) Chief cells

24) Powerful contractions that occur a few times each day in the colon are called 24) _______
A) mass movements.
B) tonic contractions.
C) segmentation.
D) phasic contractions.
E) peristalsis.

25) The swallowing center in the brain, which coordinates the muscular reflexes, is located in the 25) _______
A) medulla oblongata.
B) hypothalamus.
C) pons.
D) cerebrum.
E) cerebellum.

26) Which does NOT occur when you swallow? 26) _______
A) The lower esophageal sphincter relaxes.
B) Respiration is inhibited.
C) The upper esophageal sphincter closes.
D) The epiglottis closes.

27) Mucus functions in 27) _______
A) lubrication only.
B) protection, lubrication, and enzyme activation.
C) protection only.
D) enzyme activation only.
E) protection and lubrication.

28) In the digestive system, HCl is released by ________, whereas HCO_3^- is secreted primarily from the ________. 28) _______
A) the pancreas, parietal cells of the stomach
B) parietal cells of the stomach, pancreas
C) the liver, parietal cells of the stomach
D) parietal cells of the stomach, liver

29) Nutrient absorption occurs primarily in the 29) _______
A) large intestine.
B) stomach.
C) liver.
D) small intestine.
E) stomach and small intestine.

30) Amylases, the enzymes used to digest carbohydrates, are secreted by 30) _______
A) the pancreas into the intestine only.
B) salivary glands into the mouth and the pancreas into the intestine.
C) salivary glands into the mouth and gastric glands into the stomach.
D) gastric glands into the stomach only.
E) salivary glands into the mouth only.

31) Nearly 90% of our dietary calories from fat are in the form of 31) ________
A) steroids.
B) triglycerides.
C) cholesterol.
D) fat-soluble vitamins.
E) phospholipids.

32) Bile is 32) ________
A) made by the gallbladder only.
B) secreted by hepatocytes, made by the gallbladder, and released into the stomach.
C) released into the stomach only.
D) secreted by hepatocytes and made by the gallbladder.
E) secreted by hepatocytes only.

33) Functions of the large intestine include 33) ________
A) absorption of most products of digestion.
B) temporary food storage.
C) absorption of water and production of feces.
D) chemical digestion of chyme.
E) All of these answers are correct.

34) The release of many GI tract hormones is stimulated by a particular food or substance. Which hormone is NOT paired with its stimulus? 34) ________
A) CCK – fatty foods
B) GIP – glucose in the small intestine
C) gastrin – peptides and amino acids
D) motilin – acid in the stomach
E) secretin – acid in the small intestine

35) During the cephalic phase of gastric secretion, 35) ________
A) production of gastric juice is inhibited.
B) there are increased action potentials along the vagus nerve to the stomach.
C) secretin inhibits parietal and chief cells.
D) the intestine reflexively inhibits gastric emptying.
E) the stomach responds to distention.

36) The gastric phase of gastric secretion is triggered by the 36) ________
A) entry of food into the stomach.
B) entry of chyme into the large intestine.
C) entry of chyme into the small intestine.
D) release of cholecystokinin and secretin by the small intestine.
E) sight, thought, or smell of food.

37) An enzyme that will digest proteins into amino acids is 37) ________
A) maltase.
B) amylase.
C) nuclease.
D) lipase.
E) carboxypeptidase.

38) Most products of fat digestion are absorbed by 38) ________
A) lymphatic lacteals.
B) capillaries.
C) arterioles.
D) veins.

39) Which is true about GI muscle contractions? 39) ________
A) Tonic contractions are sustained for minutes and occur in the small intestine.
B) Contractions of the smooth muscle do not depend on calcium.
C) Phasic contractions last only seconds and occur in the stomach and small intestine.
D) Cycles of smooth muscle contraction and relaxation are associated with fast wave potentials.
E) None of these statements are true.

40) Which statement is true? 40) ________
A) Fructose moves across the apical membrane by active transport.
B) Glucose and galactose use different transporters in absorption.
C) Glucose and galactose absorption uses an apical Na^+–glucose SGLT symporter.
D) A basolateral GLUT5 transporter moves glucose out of the intestinal epithelial cell.
E) None of the statements are true.

41) Bicarbonate secretion 41) ________
A) neutralizes acid entering from the stomach into the duodenum and is secreted by apical Cl^-–HCO_3^- exchanger.
B) neutralizes acid entering from the stomach into the duodenum.
C) is not dependent on high levels of carbonic anhydrase to maintain bicarbonate production.
D) is secreted by apical Cl^-–HCO_3^- exchanger.
E) is secreted by the acinar cells.

42) Saliva secretion is primarily a result of 42) ________
A) increased sympathetic stimulation.
B) increased somatic motor stimulation.
C) increased parasympathetic stimulation.
D) decreased somatic motor stimulation.
E) decreased parasympathetic stimulation.

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the following structures with the appropriate description.

43) location of most peptic ulcers 43) ________

44) section where chyme is processed to remove water and electrolytes, leaving waste products of digestion 44) ________

45) located at the ventral end of the cecum 45) ________

46) distal–most section of small intestine 46) ________

A) duodenum
B) colon
C) ileum
D) appendix

Match the following structures with their functions.

47) chyme is released from here — 47) _______

48) organ that adds secretions to the duodenum through a duct — 48) _______

49) location of a smooth muscle band that prevents premature emptying of the stomach — 49) _______

50) organ where most digestion occurs — 50) _______

A) pylorus
B) small intestine
C) stomach
D) pancreas

Match the structure to its function.

51) Carbohydrate digestion begins here. — 51) _______

52) Carbohydrate digestion by human enzymes is completed here. — 52) _______

53) Protein digestion begins here. — 53) _______

54) Protein digestion is completed here. — 54) _______

55) Fat digestion begins here. — 55) _______

56) Fat digestion by human enzymes is completed here. — 56) _______

A) stomach
B) small intestine
C) mouth

Match each product with the cell or region that secretes or contains it.

57) parietal cells — 57) _______

58) goblet cells — 58) _______

59) brush border — 59) _______

60) pancreatic cells — 60) _______

A) enzymes
B) more than one of these
C) HCl
D) mucus

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

61) The largest collection of ________ tissue in the body is the gut-associated lymphoid tissue (GALT). 61) _______
A) lymphoid B) nervous C) epithelial D) muscle

62) Digestion is almost completely finished in the 62) _______
A) small intestine. B) large intestine. C) stomach. D) anus.

63) The three sections of the small intestine, in order according to movement of its contents, are 63) ________
A) jejunum, ileum, colon.
B) duodenum, ileum, jejunum.
C) jejunum colon, ileum.
D) duodenum, jejunum, ileum.

64) Most absorbed nutrients first enter the blood of the ________ system. 64) ________
A) lymphatic
B) immune
C) hepatic portal
D) arterial

65) The primary complex carbohydrate ingested and digested by people is 65) ________
A) starch.
B) cellulose.
C) fiber.
D) glycogen.
E) glucagon.

66) After processing in the stomach, the gastric contents are referred to as 66) ________
A) food.
B) chyme.
C) feces.
D) filtrate.

67) Bile salts aid in the digestion of fats by ________ large fat droplets. 67) ________
A) generating
B) digesting fats within
C) emulsifying
D) absorbing

68) ________ is intestinal gas produced by bacteria in the colon during the metabolism of indigestible carbohydrates. 68) ________
A) Defecation
B) Haustra
C) Flatus
D) Tenia Coli

69) The enzyme that digests starch into disaccharides is 69) ________
A) maltase.
B) sucrose.
C) amylase.
D) lactase.

70) Maltose is broken down by maltase into two molecules of 70) ________
A) glucose.
B) galactose.
C) amylase.
D) starch.

71) ________ are tiny droplets of fatty acids, cholesterol, phospholipids, mono- and diglycerides, and bile salts. 71) ________
A) Micelles
B) Lacteals
C) Enterocyte membranes
D) Chylomicrons

72) Slow waves originate in modified smooth muscle cells called 72) ________
A) mucous cells.
B) goblet cells.
C) smooth muscle sphincters.
D) interstitial cells of Cajal.

73) The ________ is a "housekeeping function" that sweeps food remnants and bacteria out of the upper GI tract and into the large intestine. 73) ________
A) slow wave
B) segmental contraction
C) migrating motor complex
D) peristaltic contraction

74) ________ involve short segments of intestine that alternately contract and relax. They are responsible for ________. 74) ________
A) Peristaltic contractions, mixing
B) Segmental contractions, mixing
C) Segmental contractions, pushing a bolus forward
D) Peristaltic contractions, pushing a bolus forward

75) The exocrine portion of the pancreas consists of lobules called ________, which secrete ________. 75) ________
A) islets, hormones
B) islets, digestive enzymes
C) acini, digestive enzymes
D) acini, hormones

76) ________ digest terminal peptide bonds to release amino acids. 76) ________
A) Proteases
B) Lipases
C) Endopeptidases
D) Exopeptidases

77) Bile is secreted from the ________ and stored in the ________. 77) ________
A) liver, gallbladder
B) hepatocytes, pancreas
C) pancreas, hepatocytes
D) gallbladder, liver

78) Digestive reflexes integrated in the CNS are called ________ reflexes. 78) ________
A) long
B) deglutition
C) short
D) defecation

79) Long reflexes that originate outside the digestive system include ________ reflexes and ________ reflexes, which are called ________ reflexes. 79) ________
A) feedforward, emotional, short
B) submucosal plexus, myenteric plexus, short
C) submucosal plexus, myenteric plexus, cephalic
D) feedforward, emotional, cephalic

80) Short reflexes of the digestive system are integrated in the ________ nervous system. 80) ________
A) parasympathetic
B) enteric
C) central
D) peripheral

81) The primary products of protein digestion are ________, ________, and ________. 81) ________
A) polysaccharides, disaccharides, monosaccharides
B) nucleic acid polymers, DNA, RNA
C) free amino acids, dipeptides, tripeptides
D) triglycerides, fatty acids, glycerol

82) Enzymatic digestion of fats involves ________, which breaks down ________. 82) ________
A) bile, cholesterol
B) pepsin, proteins
C) lipase, triglycerides
D) amylase, carbohydrates

83) ________ is a protein cofactor secreted by the pancreas that allows lipases access to fats inside the bile salt coating. 83) ________
A) Lipase
B) Phospholipase
C) Colipase
D) Monoglyceride

84) Vitamin ________ must be complexed with a protein called ________ to be absorbed from the small intestine. 84) ________
A) K, hydrogen sulfide
B) B_{12}, intrinsic factor
C) A, Olestra
D) D, somatomedins

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

85) Defects in which channel lead to the disease cystic fibrosis? Which ion does this channel transport?

86) What is the ENS, and when was it discovered? What is its significance, what is its nickname, and what are the characteristics that justify that nickname?

87) Where are long reflexes and short reflexes integrated?

88) In one sentence, briefly summarize the generalized function of regulatory peptides secreted by the GI tract. Then, list the key GI hormones, spelling them out and also writing their abbreviations where appropriate.

89) List the six types of epithelial cells associated with gastric glands and the secretions from each.

90) Name two functions of cholecystokinin (CCK).

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

91) Put the following steps of fat digestion and absorption in order: 91) _______

1. Bile salts coat fat droplets.
2. Cholesterol is transported into cells.
3. Chylomicrons are formed.
4. Chylomicrons are removed by lymphatic system.
5. Monoglycerides and fatty acids move out of micelles.
6. Pancreatic lipase and colipase break down fats.

A) 2, 6, 5, 1, 3, 4
B) 1, 6, 5, 2, 3, 4
C) 1, 2, 3, 4, 6, 5
D) 6, 3, 4, 1, 2, 5
E) None of the answers are correct.

92) During defecation, 92) _______

A) the external anal sphincter contracts.
B) the internal anal sphincter contracts.
C) distension in the rectal wall activates a spinal reflex.
D) the internal anal sphincter is consciously relaxed.
E) distension in the rectal wall activates a short reflex.

93) Decreased levels of bile salts in the bile would interfere with digestion of 93) _______

A) protein.
B) nucleic acids.
C) fat.
D) carbohydrates.
E) vitamins and minerals.

94) In response to the acid in the duodenum, the blood levels of 94) _______

A) gastrin increase.
B) secretin increase.
C) cholecystokinin decrease.
D) enterocrinin decrease.
E) All of these answers are correct.

95) A blockage of the ducts from the parotid glands would 95) _______
A) result in the production of less viscous saliva.
B) decrease carbohydrate digestion in the mouth.
C) decrease the lubricating properties of saliva.
D) decrease the sense of taste.
E) cause all of these effects.

96) In response to the hormone secretin, the pancreas secretes a fluid that contains 96) _______
A) only amylase.
B) bile.
C) only proteases.
D) bicarbonate.
E) enzymes.

97) In response to the hormone cholecystokinin (CCK), the pancreas secretes a fluid that contains 97) _______
A) only proteases.
B) bicarbonate.
C) only amylase.
D) bile.
E) enzymes.

98) Digestion of an unlabeled carbohydrate results in increased amounts of the monosaccharides glucose and galactose. Which is most likely to be the original, unlabeled carbohydrate? 98) _______
A) sucrose B) glycogen C) maltose D) lactose E) cellulose

99) Diarrhea often results from intestinal infections. Why? 99) _______
A) Dehydration of the body will kill the bacteria causing the infection.
B) Loss of electrolytes will kill bacteria.
C) Bacterial toxins enhance the secretion of Cl^- and water.
D) The immune system increases the production of enzymes for added protection.

100) A drug that blocks the action of carbonic anhydrase in parietal cells would result in 100) ______
A) increased protein digestion in the stomach.
B) decreased production of pepsinogen by chief cells.
C) decreased gastrin production.
D) a lower pH during gastric digestion.
E) a higher pH during gastric digestion.

101) A drug that blocks the action of the hormone cholecystokinin would affect 101) ______
A) the amount of bile produced by the liver.
B) the level of intestinal gastrin.
C) pancreatic secretions.
D) secretions of the duodenal glands.
E) All of the answers are correct.

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the hormone with the correct statement.

102) secreted by cells in the stomach	A) gastrin	102) ______
103) inhibits gastric emptying	B) motilin	103) ______
104) stimulates bile release	C) gastric inhibitory peptide	104) ______
105) stimulates insulin release	D) secretin	105) ______
106) smooth muscle of intestines is a target	E) cholecystokinin	106) ______

Match the structure or compound to its function.

107) apical transporter for iron absorption	A) ezetimibe	107) ______
108) basolateral transporter for ionized iron	B) DMT1	108) ______
109) hormone that decreases iron absorption	C) hepcidin	109) ______
	D) NPC1L1	
110) blocks cholesterol absorption	E) ferroportin	110) ______
111) transporter for cholesterol absorption		111) ______

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

112) Where does carbohydrate digestion begin? Which enzyme is involved, and which reaction does it catalyze?

113) The mucosa of the majority of the digestive tract contains simple columnar epithelium, while the esophagus and the anus have stratified squamous epithelium. Why is this the case? What purpose does each type of epithelium serve in the digestive tract?

114) List the key components of bile. Is bile action similar to the action of lipases? Explain.

115) List the four basic processes of the digestive system, and define each.

116) Proteins must first be enzymatically degraded to single amino acids before entering the capillaries of the hepatic portal system. Is this true or false? What is the significance of this? Is absorption of carbohydrates and lipids restricted to monomers? Explain.

117) Draw a cell that secretes bicarbonate into the lumen of the pancreas, selecting transporters and/or enzymes from following list (you may or may not need them all):

chloride channel
CFTR channel
GLUT4
H^+–K^+–ATPase
Na^+–Cl^-–K^+ Symport
Cl^- – HCO_3^- Antiport
Cl^- – HCO_3^- Symport
carbonic anhydrase
K^+ leak channel
leaky junctions
Na^+–K^+–ATPase
Na^+–H^+ antiport

Label the cell showing the lumen, interstitial fluid, the basolateral membrane, and the apical membrane.

118) Compare and contrast gastrin, CCK, secretin, GIP, motilin, and glucagon–like peptide 1. In your answer, include the following details for each term: *secreted by, target(s), effects,* and *stimulus for release.*

119) Draw a map of a short reflex that might occur in response to ingestion of food. Include on your map the stimuli, afferent sensors, integrating centers, efferent pathways, targets, and tissue responses.

120) Diagram intestinal absorption of peptides. Begin with the arrival of a protein in the lumen of the small intestine, and continue the diagram until the absorbed peptides reach the hepatic portal vein. Include in your diagram the enzymes and transporters that might be involved; and label the intestinal lumen, the apical and basolateral membranes the peptides might cross, and the anatomical pathway the absorbed peptides would take to reach the hepatic portal vein.

121) Make a map showing the relationships of the following terms involved with gastric secretion.

chief cell
D cell
ECL cell
enteric sensory neuron
G Cell
gastrin
H^+
histamine
parietal cell
pepsin
pepsinogen
somatostatin
vagus nerve

Where appropriate, indicate stimulatory or inhibitory effect. Terms may be added as needed.

122) Name the proenzymes secreted by the pancreas. Which one is involved in the activation of the other proenzymes? What are their active forms?

123) The enteric nervous system and the CNS have several similarities. What are they?

124) Draw a cell that secretes HCl into the lumen of the stomach, selecting transporters and/or enzymes from the follc list (you may or may not need them all):

chloride channel
CFTR channel
GLUT4
carbonic anhydrase
H^+–K^+–ATPase
Cl^- – HCO_3^- Antiport
Cl^- – HCO_3^- Symport
Na^+–K^+–ATPase

Label the cell showing the lumen, interstitial fluid, the basolateral membrane, and the apical membrane.

125) One way to determine the total energy in calories of a food is to burn it in a device called a bomb calorimeter. The heat produced during combustion minus the heat added to begin the combustion is the total calories in the food. The typical human digestive tract does not absorb all of the calories in food; thus it is not 100% efficient. Also, calories in indigestible organic compounds may not be absorbed at all, and thus can be considered to be "ze calorie" relative to human nutrition. Artificial sweeteners and fats are designed to produce desired sensation in tl mouth without being absorbed. These are organic compounds that would release heat in a bomb calorimeter, thu technically have calories.
A. What does this suggest about the accuracy of the bomb calorimeter in determining available food calories for humans? What would be the results of bomb calorimetry of human feces?
B. What is it about a particular food substance that may prevent absorption of its calories?
C. What happens to the "zero-calorie" molecules as they pass through the digestive tract? What are the potential effects on defecation?

126) Cody has always carefully controlled her caloric intake, and is lean. During her pregnancy, she continued to eat the same, hoping that taking prenatal vitamins and minerals would help her fetus would grow normally. Her only complication during pregnancy was persistent constipation and hemorrhoids. She gained only the minimum weight suggested by her doctor, and delivered a healthy, full-term infant. She quickly returned to her pre-pregnancy weight. Propose an explanation for how she managed to gain enough weight, did not violate the law of conservation of mass, and for her only complication. What does this suggest about "eating for two" that some pregnant women do?

127) A condition known as lactose intolerance is characterized by painful abdominal cramping, gas, and diarrhea. The cause of the problem is an inability to digest the milk sugar, lactose. How would this cause the observed symptoms?

128) Muna came down with the flu. She experienced severe vomiting or mostly stomach contents, for three days, and now is having chest pains. She is calling one of her symptoms "heartburn," but reported that it felt as if she had swallowed a small apple. Certain that she was having a heart attack, Muna rushed to the emergency department. The doctor took her history and symptoms, and reassured her that she only had esophagitis. How did the physician conclude this? Is her blood pH higher or lower than normal?

129) In some severe cases, a person suffering from stomach ulcers may have surgery to cut the branches of the vagus nerve that innervates the stomach. How would this help the problem?

130) Essential nutrients are those that our cells require but cannot make, and thus they must be present in the diet. Only some amino acids and some fatty acids are essential. What does that suggest about extreme diets that eliminate fats or proteins? What does that suggest about our carbohydrate intake? Describe the typical American diet, in terms of relative amounts of ingested carbohydrates, fats, proteins, and nucleic acids. Propose some explanations for why one of those is predominant.

131) What would be the nutritional consequence of excessive antacid use?

132) You and your lab partners in a human anatomy course removed the intestines from an adult cadaver, and cut away connective tissues as necessary to uncoil the intestines into a straight line that could be measured. Your professor told you that the intestinal tract of an adult of this size and gender is typically about 13 ft. Your measurement is 22 ft. One of the lab partners insists that everything is smaller in preserved cadavers due to dehydration. Address this suggestion, and propose some other explanations for the difference.

133) You are studying the toxic effects of a newly-discovered tropical plant extract. From preliminary studies, it appears that the toxin prevents H^+-dependent membrane transport. Explain which digestive processes may be impaired.

134) Cholesterol is absorbed without being digested into smaller pieces. How does this compare to absorption of carbohydrates and proteins? Which characteristic of cholesterol suggests it would be transported by simple diffusion? What is the evidence that transport proteins are involved in cholesterol absorption? Does this discovery rule out transport by simple diffusion? Explain.

135) Chronic inflammation of the pancreas, or pancreatitis, impairs the digestive functions of this organ. What would be the effect on digestion and absorption of carbohydrates, fats, and proteins if acinar cells are impaired? What if duct cells are impaired? Which hormones may be used to treat pancreatitis?

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

136) In the body, 80% of all lymphocytes, a type of immune system cell, are thought to be present in the 136) ______
A) mouth.
B) large intestine.
C) stomach.
D) small intestine.
E) appendix.

137) Feedforward reflexes are also ________ reflexes. 137) ______
A) cephalic B) short C) emotional D) medium

138) A possible treatment for *Salmonella* could involve 138) ______
A) activating cytokine receptors. B) increasing macrophages.
C) increasing lymphocytes. D) blocking M cell receptors.

139) Vomiting and diarrhea are similar since both 139) ______
A) are used to prepare patients for surgery.
B) work to rid the body of damaging agents.
C) are stimulated by the CNS.
D) use normal GI motility to move substances.

Exam

Name___________________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Adipocytes secrete the hormone 1) ________
A) leptin.
B) insulin.
C) neuropeptide Y.
D) ghrelin.
E) orexin.

2) Most obese humans are deficient in leptin. 2) ________
A) True B) False

3) The brain neurotransmitter that is the stimulus for food intake is 3) ________
A) orexin.
B) ghrelin.
C) neuropeptide Y.
D) leptin.
E) insulin.

4) Energy input equals energy output. This statement reflects the 4) ________
A) second law of thermodynamics.
B) first law of thermodynamics.
C) law of mass balance.

5) The first law of thermodynamics states that 5) ________
A) natural processes move from order to disorder, or entropy.
B) the total amount of energy in the universe is constant.
C) energy can be created or destroyed.
D) what heats up must cool down.
E) there is no way a process can increase the amount of order.

6) One kilocalorie (kcal) is the amount of energy needed to raise 1 ________ of water by one degree Celsius. 6) ________
A) tablespoon
B) gallon
C) cup
D) milliliter
E) liter

7) A bomb calorimeter measures 7) ________
A) the carbon dioxide produced when a food sample is completely combusted.
B) both carbon dioxide produced and oxygen consumed when a food sample is completely combusted.
C) the heat released when a food sample is burned.
D) only the food calories available to the human body.
E) the oxygen consumed when a food sample is completely combusted.

8) Which can be measured to determine a person's metabolic rate? 8) ________
A) only oxygen consumed by body
B) only carbon dioxide produced by body
C) only heat released from body
D) heat released from body, oxygen consumed by body, and carbon dioxide produced by body
E) None of these answers are correct.

9) The respiratory quotient is ________ for consumed carbohydrates compared to proteins and fats. 9) ________
A) lowest B) the same C) highest

10) The most practical way to estimate a person's basal metabolic rate is to measure it when the person is 10) ________
A) resting after a 12-hour fast.
B) sleeping.
C) resting after a large meal.

11) *Metabolism* is a term that describes 11) ________
A) only chemical reactions that release ATP from living cells.
B) all work done by a living organism.
C) the energy released from chemical bonds in living cells.
D) all chemical reactions that take place within an organism.
E) the extraction of nutrients from biomolecules.

12) *Anabolic pathways include* 12) ________
A) reactions that release energy and reactions that result in the synthesis of large biomolecules.
B) reactions that require a net input of energy.
C) reactions that release energy.
D) reactions that result in the synthesis of large biomolecules.
E) reactions that require a net input of energy and reactions that result in the synthesis of large biomolecules.

13) The brain relies solely on ________ as its energy source. If it is deprived of this substrate, the cells soon die. 13) ________
A) glucose B) amino acids C) fatty acids D) insulin

14) Gluconeogenesis refers to 14) ________
A) converting glucose to glycogen storage.
B) removing fatty acids from adipose storage.
C) creating glucose from non-carbohydrate precursors.
D) converting glucose to storage as adipose tissue.
E) removing glucose from storage as glycogen.

15) An enzyme that digests fats is 15) ________
A) kinase. B) hydrolase. C) lipase. D) ligase. E) lyase.

16) The liver and skeletal muscles store glucose as ________ for a ready energy source. 16) _______
A) glycogen
B) glucose
C) urea
D) ketone bodies
E) lipids

17) Which is NOT produced in the body for the purpose of storing extra calories? 17) _______
A) protein
B) carbohydrate
C) fat
D) All of these are produced in the body for that purpose.

18) The nutrients that yield the most energy per gram when metabolized are 18) _______
A) fats.
B) carbohydrates.
C) nucleic acids.
D) proteins.
E) vitamins.

19) Diet-induced thermogenesis is highest after ingestion of 19) _______
A) proteins.
B) carbohydrates.
C) fats.
D) spicy foods, like jalapeno peppers.

20) The synthesis of glucose from a noncarbohydrate precursor is referred to as 20) _______
A) gluconeogenesis.
B) glycogen.
C) glycogenolysis.
D) glycolysis.
E) glycogenesis.

21) During the fasting state, the energy stores of the ________ become the major source of glucose for the whole body. 21) _______
A) brain
B) muscles
C) liver
D) adipose tissues
E) pancreas

22) Lipoproteins that carry mostly cholesterol and phospholipids out of the plasma to the liver are called 22) _______
A) very low-density lipoproteins (VLDLs).
B) low-density lipoproteins (LDLs).
C) very high-density lipoproteins (VHDLs).
D) high-density lipoproteins (HDLs).
E) intermediate-density lipoproteins (IDLs).

23) Before converting amino acids into intermediates for energy metabolism, they must first undergo 23) ________
A) delousing.
B) denaturation.
C) depeptidization.
D) deamination.
E) detoxification.

24) When amino acids are metabolized for energy, the nitrogen is converted to urea and then is excreted from the body by the 24) ________
A) sweat glands.
B) spleen.
C) skin.
D) kidneys.
E) liver.

25) The reactions where fats are broken down into glycerol and fatty acids are called 25) ________
A) gluconeogenesis.
B) glycolysis.
C) lipolysis.
D) glycerolysis.
E) liposuction.

26) The process of disassembling fatty acids into two-carbon units inside mitochondria is called 26) ________
A) deamination.
B) ketonization.
C) beta-oxidation.
D) oxidative phosphorylation.
E) chemiosmosis.

27) A substance that the liver produces when metabolizing fatty acids that is both an energy source and a potentially-harmful substance is 27) ________
A) beta units.
B) ammonia.
C) rancid fatty acids.
D) urea.
E) ketone bodies.

28) The alpha cells of the pancreas secrete 28) ________
A) digestive enzymes.
B) cortisol.
C) glucagon.
D) renin.
E) insulin.

29) The beta cells of the pancreas produce 29) ________
A) insulin.
B) renin.
C) digestive enzymes.
D) cortisol.
E) glucagon.

30) D cells in the islet of Langerhans secrete 30) _______
A) amylin.
B) glucagon.
C) pancreatic polypeptide.
D) somatostatin.
E) insulin.

31) When blood glucose levels increase, as in the "fed" state, 31) _______
A) only peripheral cells take up less glucose.
B) only insulin is secreted.
C) only protein synthesis decreases.
D) only glucagon is secreted.
E) All of these answers are correct.

32) When blood glucose levels decrease, as in the "fasted" state, only 32) _______
A) protein synthesis decreases.
B) peripheral cells take up less glucose.
C) insulin is secreted.
D) glucagon is secreted.
E) All of these answers are correct.

33) The feeding and satiety centers are located in the 33) _______
A) cerebrum.
B) cerebellum.
C) medulla oblongata.
D) pons.
E) hypothalamus.

34) During the absorptive state, 34) _______
A) the liver synthesizes glycogen.
B) adipocytes release fatty acids to the circulation.
C) glucagon levels are increased.
D) skeletal muscles break down glycogen.
E) All of these answers are correct.

35) During starvation, 35) _______
A) structural proteins cannot be used as a potential energy source.
B) gluconeogenesis increases.
C) circulating ketone bodies decrease.
D) carbohydrate utilization increases.
E) All of these answers are correct.

36) The brain can use ________ for energy. 36) _______
A) only fats
B) only glucose
C) only ketones
D) only lactate
E) both glucose and ketones

37) Which factors increase basal metabolic rate? 37) _______
A) acetylcholine
B) thyroid hormones and epinephrine
C) epinephrine
D) insulin
E) thyroid hormones

38) Insulin secretion 38) _______
A) is inhibited by GLP-1.
B) is stimulated by parasympathetic neurons.
C) decreases in response to increased amino acid concentrations.
D) is stimulated by sympathetic neurons.
E) None of these answers are correct.

39) A primary target tissue for insulin is the 39) _______
A) intestine.
B) liver.
C) brain.
D) brain and liver.
E) brain, liver, and intestine.

40) GLUT4 transporters are 40) _______
A) stored in cytoplasmic vesicles and found in adipose and skeletal muscles.
B) stored in cytoplasmic vesicles.
C) found in adipose and skeletal muscles.
D) inserted into the plasma membrane by endocytosis.
E) inserted into the plasma membrane in response to glucagon.

41) Insulin 41) _______
A) only stimulates glycolysis.
B) stimulates glycolysis and inhibits gluconeogenesis.
C) only stimulates lipolysis.
D) stimulates glycolysis and lipolysis.
E) only inhibits gluconeogenesis.

42) Glucagon 42) _______
A) stimulates gluconeogenesis and primarily targets the liver.
B) stimulates gluconeogenesis.
C) primarily targets the liver.
D) stimulates gluconeogenesis and primarily targets skeletal muscle.
E) primarily targets skeletal muscle.

43) Dehydration can result from 43) _______
A) heat exhaustion and diabetes.
B) diabetes.
C) gluconeogenesis.
D) heat exhaustion.
E) CCK.

44) Amylin 44) _______
A) helps regulate glucose homeostasis by speeding up gastric emptying.
B) is co-secreted with insulin.
C) is co-secreted with glucagon.
D) is co-secreted with insulin and helps regulate glucose homeostasis by speeding up gastric emptying.
E) is co-secreted with glucagon and helps regulate glucose homeostasis by speeding up gastric emptying.

45) In type I diabetes, a hyperglycemic hyperosmolar state may occur. Which best describes this state? 45) _______
A) Blood osmolarity is below normal levels due to elevated glucose.
B) Plasma glucose and blood osmolarity levels are above normal.
C) There is decreased water intake.
D) There are low levels of vasopressin (ADH).
E) ATP production is increased due to the increased levels of glucose in cells.

46) Homeothermic refers to 46) _______
A) regulating body temperature within a narrow range.
B) natural treatments or remedies.
C) shivering.
D) warm relationships with others of the same gender.

47) Convective heat loss occurs when 47) _______
A) a cooler object rests on the body's surface.
B) one swims in water below body temperature.
C) warm air rises from the body's surface.
D) water evaporates from the skin's surface.
E) All of the answers are correct.

48) Nearly half of heat is lost from the body through the process of 48) _______
A) conductive heat loss.
B) convective heat loss
C) evaporative heat loss.
D) radiant heat loss.
E) concentrative heat loss.

49) Specific responses to changes in body temperature are regulated by the 49) _______
A) hypothalamus.
B) cardiac output.
C) thermoreceptors.
D) medulla oblongata.
E) skin.

50) Heat loss is promoted by 50) _______
A) sweating, dilation of cutaneous blood vessels, and nonshivering thermogenesis.
B) dilation of cutaneous blood vessels.
C) sweating and dilation of cutaneous blood vessels.
D) nonshivering thermogenesis.
E) sweating.

51) Prediabetes is a condition that will likely become diabetes if eating and exercise habits are not altered. 51) _______
A) True
B) False

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Classify each scenario below as to primarily which type of work is being done.

52) maintaining a concentration gradient across a membrane 52) _______
53) working on a dock, loading and unloading boxes into trucks all day 53) _______
54) wound healing, recovering from surgery 54) _______
55) enlarging one's muscles through body–building exercises 55) _______
56) bringing glucose molecules inside brain cells so one can think 56) _______
57) a thoroughbred horse, or greyhound dog, running around the track 57) _______

A) mechanical work
B) transport work
C) chemical work

Match each term to its definition.

58) a series of interconnected chemical reactions 58) _______
59) reactions that result in the breakdown of large biomolecules 59) _______
60) reactions that result in the synthesis of large molecules 60) _______

A) anabolic pathway
B) biochemical pathway
C) catabolic pathway

Match the nutrient to its primary fate.

61) used immediately for energy 61) _______
62) synthesis of tissues 62) _______
63) storage 63) _______

A) carbohydrate
B) fat
C) protein

Match the term to its definition.

64) the sum of all the body's chemical reactions — 64) _______

65) large molecules synthesized from smaller ones — 65) _______

66) large molecules broken into smaller ones — 66) _______

A) catabolism

B) anabolism

C) metabolism

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

67) The energy content of a food or similar substance is measured by a process called ________. 67) ______________

68) The ________ represents the minimum resting energy expenditures of an awake, alert individual. 68) ______________

69) Measuring oxygen consumption as a way of estimating a person's metabolic rate is one form of ________. 69) ______________

70) A diet of pure carbohydrate would produce an RQ of ________. 70) ______________

71) Metabolism is often divided into ________, energy-producing reactions, and ________, energy-utilizing reactions. 71) ______________

72) The enzyme ________ converts triglycerides into free fatty acids and glycerol. 72) ______________

73) The lipoprotein most likely to be called "good (or healthy) cholesterol" is ________. 73) ______________

74) ________ are lipoprotein and lipid complexes that are formed in the intestine to carry lipids into circulation. 74) ______________

75) Lipoproteins that contain large amounts of cholesterol for transport to peripheral tissues are called ________. 75) ______________

76) ________ are unstable molecules that are formed during normal metabolism and are thought to contribute to aging, disease, and some cancers. 76) ______________

77) ________ are molecules that prevent damage to our cells by stopping the destructive effects of free radical formation. 77) ______________

78) Triglycerides are broken down into ________, which feeds into glycolysis, and ________, which are metabolized to acetyl CoA. 78) ______________

79) The pancreatic hormone that increases blood glucose concentration is ________. 79) ______________

80) The process of synthesizing glucose from noncarbohydrate precursors such as amino acids is called ________. 80) ______________

81) The process of glycogen synthesis is known as ________. 81) ______________

82) ________ is the amount of heat generated during the digestion of a meal. 82) ______________

83) ________ is a condition of insulin deficiency from beta cell destruction. 83) ______________

84) ________ is known as insulin-resistant diabetes. 84) ______________

85) The loss of water in the urine due to unreabsorbed solutes is known as ________. 85) ______________

86) ________ is a test used to measure the body's insulin response to an ingested glucose load. 86) ______________

87) The combination of type 2 diabetes, atherosclerosis, and high blood pressure is called ________. 87) ______________

88) ________ monitor skin temperature and core body temperature. 88) ______________

89) The production of heat from rhythmic tremors of skeletal muscle is referred to as ________. 89) ______________

90) Chemicals known as ________ are fever-producing cytokines that are part of the normal immune response. 90) ______________

91) A genetic condition in which body temperature becomes abnormally elevated is ________. 91) ______________

92) ________ is the direct transfer of heat energy from one object to another. 92) ______________

93) ________ is the transfer of heat energy to air. 93) ______________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

94) After ingestion of carbohydrates in an insulin-deficient diabetic, ________ would happen. 94) _______
A) ketone production
B) glycolysis
C) fat synthesis
D) protein breakdown
E) protein breakdown and ketone production

95) Drugs used to treat diabetes may 95) _______
A) stimulate beta-cell secretion of insulin.
B) stimulate digestion and absorption of carbohydrates in the intestine.
C) decrease responsiveness of target tissue to insulin.
D) stimulate hepatic glucose output.
E) None of these answers are correct.

96) In the fasted state, which statement is FALSE? 96) _______
A) Some amino acids will be deaminated.
B) Adipose breaks down its store of triglycerides.
C) Skeletal muscle will increase gluconeogenesis.
D) Pyruvate and lactate are transported to the liver to make glucose.

97) The level of ketone bodies in the blood increases when high levels of ________ are being metabolized. 97) _______
A) carbohydrates
B) glucose
C) amino acids
D) fatty acids
E) proteins

98) The Atkins and South Beach diets are considered ketogenic because 98) _______
A) Type I diabetics use them.
B) they have the potential to increase blood pH.
C) they cause the body to burn calories from noncarbohydrate sources.
D) they require people to eat large quantities of ketones.
E) they shift metabolism to increased glycolysis.

99) When blood levels of glucose, amino acids, and insulin are high, and glycogenesis is occurring in the liver, the body is in the 99) _______
A) stress state.
B) bulimic state.
C) postabsorptive state.
D) absorptive state.
E) fasted state.

100) If you were in a desert without a food source, which nutrient would you like to have stored in your body in a large amount? 100) ______
A) glycogen
B) vitamin B_6
C) protein
D) fat
E) calcium

101) Which symptom would you expect to observe in a person with type 1 diabetes mellitus? 101) ______
A) ketoacidosis
B) glucosuria
C) hyperglycemia
D) thirst and polydipsia
E) All of these answers are correct.

102) In type 2 diabetes mellitus, insulin levels are frequently normal, yet the target cells are less sensitive to the effects of insulin. This suggests that the target cells 102) ______
A) have adequate internal supplies of glucose.
B) are impermeable to insulin.
C) cannot convert insulin to an active form.
D) may have a problem in their signal transduction pathway.
E) None of these answers are correct.

103) Which drug would be an effective way to lower plasma cholesterol levels? 103) ______
A) a drug that stimulates intestinal cholesterol transport
B) a drug that adds plant sterols and stanols to the diet
C) a drug that binds to bile acids and prevents them from being reabsorbed
D) a drug that binds to bile acids and prevents them from being reabsorbed or a drug that adds plant sterols and stanols to the diet
E) a drug that stimulates the enzyme HMG coA reductase

104) Two males are the same age and ethnic background. While being tested for their BMR, Ibrahim consumes 20 liters of oxygen/hour and Yusuf consumes 16 liters of oxygen/hour. Which of the two needs to consume the most calories in order to maintain proper health and constant weight? 104) ______
A) Ibrahim
B) Yusuf

105) On a tour of African countries, Dom contracts a bad case of traveler's diarrhea. Because he cannot eat very much, his body starts to use energy sources other than carbohydrates. This would result in 105) ______
A) ketosis.
B) increased gluconeogenesis in the liver.
C) decreased blood pH.
D) increased levels of urea in the blood.
E) All of these answers are correct.

106) Both insulin and glucagon are peptide hormones that target liver cells. The responses of the target cells to these two hormones are the opposite of each other. This information implies that 106) ______
A) both hormones interact with receptors at the cell nucleus.
B) one of the hormones does not interact with a membrane receptor.
C) each of the two hormones uses a different second messenger.
D) one hormone binds to a receptor on the cell membrane and the other to an intracellular receptor.

107) Brown fat 107) ______
A) functions in nonshivering thermogenesis.
B) contains a rich vascular supply.
C) is innervated by the sympathetic nervous system.
D) is found in infants.
E) All of these answers are correct.

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

108) List and discuss two different theories about the regulation of food intake and the role of signal molecules and the brain.

109) In the study by Morewedge et al., subjects that imagined eating 30 M&M's one at a time ate fewer real M&M's than subjects who thought about eating only 3 M&M's. What possible reason can you think of to explain this observation?

110) Explain why glucose should NOT be the sole source of energy in our diets. What is the best approach to meeting energy needs?

111) Explain the relationship among chylomicrons, cholesterol, and the various forms of lipoproteins (LDL-C, HDL-C, etc.). What is the role of each one in the body? Why are some considered "bad" and others "good"? Where does each come from? Where could each end up?

112) It is known that exercise is good for diabetics. Explain how GLUT4 transporters may be involved in this beneficial effect of exercise.

113) Compare and contrast the two types of diabetes mellitus.

114) Compare and contrast insulin and glucagon.

115) Why does insulin deficiency lead to metabolic acidosis?

116) Explain how and why abnormal temperatures, either too high or too low, can be dangerous.

117) Define basal metabolic rate (BMR), and list the six factors affecting BMR.

118) The weather person tells us that the temperature is 95, but the heat index is 103 and a heat advisory is in effect. What does this mean? Is humidity high or low? Why does it matter? Who is most affected, and why? What are some steps one could take to adjust?

119) Marta suffers from anorexia nervosa. One afternoon she is rushed to the emergency room because of cardiac arrhythmias. Her breath has the smell of an aromatic hydrocarbon (sweet, fruity odor), and blood and urine samples indicated ketonemia and ketonuria. Why do you think she is having the arrhythmias?

120) The crew of an ocean-going racing sailboat must adhere to strict weight limits for both their bodies and the food they eat, yet their activity levels are very high, and weather conditions may vary to extreme cold, so they must consume many kilocalories. What kinds of nutrient-dense foods can you suggest to them to bring? In general, what type of energy nutrient should they choose to consume?

121) Damien, in training for the Boston marathon, finds his plasma cholesterol concentration is somewhat increased. What other test results should Damien review before changing his training regime or diet?

122) Salazar has insulin-dependent diabetes mellitus. Salazar has an accident on his bike and breaks his arm. Would his insulin dosage change?

123) Haile has a blood test that shows a normal level of LDLs but an elevated level of HDLs in his blood. Since his family has a history of cardiovascular disease, he wonders if he should modify his lifestyle. What would you tell him?

124) Low leptin levels have been observed in athletes with chronic negative energy balance. What might be a possible explanation for this? Would you expect neuropeptide Y (NPY) levels to be high? Explain.

125) Describe a ketogenic diet. What are the pros and cons of such a diet? How is ketosis related to diabetes mellitus?

126) High protein (low carbohydrate) diets have become popular in recent years. Based on what you have learned about metabolism and the body's energy needs, explain what might be some of the disadvantages of this type of diet.

127) Kadija and Sally have been best friends since high school. Each is 5'5" tall and weighs about 160 lbs., which is considered to be overweight. Kadija's blood pressure is 125/82 and her blood analysis shows healthy levels of glucose and lipids. Sally's blood pressure is 135/87 and her blood analysis shows elevated glucose and triglycerides. Given that they are the same height and weight, would you expect their blood analyses to be more similar? Explain. Would you expect their body shapes to be similar? Explain. Which one is more likely to be diagnosed with metabolic syndrome? Explain.

128) There is only one way to lose weight without actually removing a body part: consume fewer calories than you expend. Is this statement true or false? Explain, citing a fundamental law of physics. Why is it difficult for many people to lose weight? Many weight-loss methods work for a short time. Explain why they work, and why they may not work forever.

129) Describe anorexia nervosa and diabetes mellitus, and how they resemble each other.

130) Prior to the discovery of the role of insulin in metabolism and the development of insulin therapy to treat diabetes mellitus, a diagnosis of this disease was always fatal. What do untreated diabetics die from?

131) Misty is 16 and, with her family, is visiting her grandmother for Christmas. At the end of a week surrounded by tempting holiday food, Misty weighs herself and sees that she has gained five pounds. She confides to her mother that she ate all of a 1-lb box of chocolates on Christmas day, but didn't overeat in any other way. There are 500 kilocalories per 100 g of chocolate. 1 lb = 454 g. There are 3500 kilocalories in a pound of fat. How many calories did she consume? Does this account for the five pounds she gained? Explain.

132) Calculate the body mass index of a woman who is 5'5" tall and weighs 180 lbs. Now calculate the BMI of a man with those dimensions. 1 kg = 2.2 lbs, 1 m = 39.24 in. Is either individual considered obese? Explain. Could BMI be misleading as a means of indicating obesity? Explain.